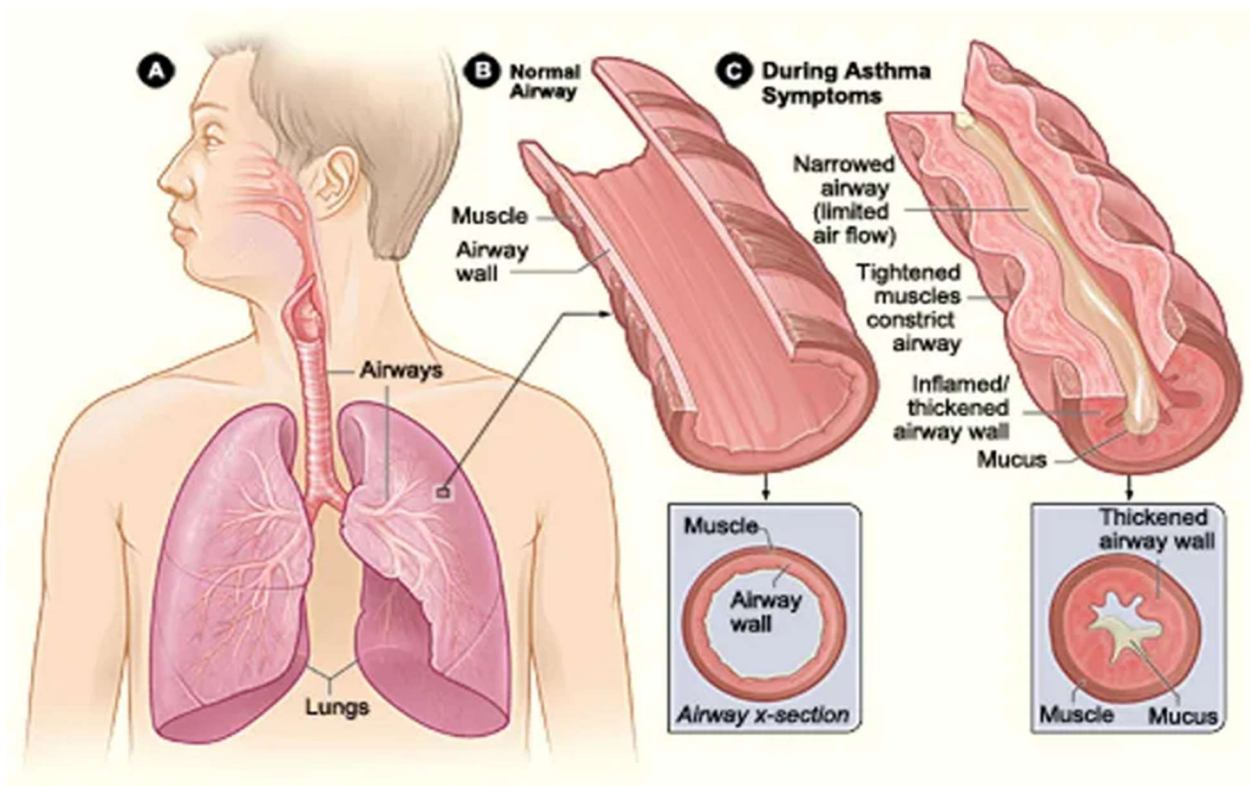


ASTHMA

In an asthma attack, the muscles of the air passages in the lungs go into spasm. As a result, the airways become narrowed, which makes breathing difficult. Sometimes, there is a recognized trigger for an attack, such as an allergy, a cold, a particular drug, or cigarette smoke. At other times, there is no obvious trigger. Many sufferers have sudden attacks. People with asthma may be treated, depending on the severity of their condition, with rescue inhalers or nebulizers on a regular basis. They can usually deal with their own attacks by using their inhaler at the first sign of an attack, but may need help and encouragement.



Recognition

- Difficulty breathing.
- Wheezing.
- Difficulty speaking, leading to short sentences and whispering.
- Coughing.
- Distress and anxiety.

- Features of hypoxia, such as a gray-blue tinge to the lips, earlobes, and nailbeds (cyanosis).
- Exhaustion in a severe attack. If the attack worsens, the casualty may stop breathing and lose consciousness.

Aim

- To ease breathing.
- To obtain medical help if necessary.

What to do;

- Keep calm and reassure the casualty. Get her to take her usual dose of her inhaler; use a spacer if she has one. Ask her to breathe slowly and deeply.
- Sit her down in the position she finds most comfortable; do not let her lie down.
- A mild attack should ease in a few minutes. If it does not, ask the casualty to take another dose from her inhaler.
- Call for emergency help if the attack is severe and any of the following occur: the inhaler has no effect; the casualty is getting worse; breathlessness makes talking hard; she becomes exhausted.
- Help the casualty use her inhaler as required. Monitor her vital signs—level of response, breathing, and pulse until help arrives.

FIRES AND BURNS

Fire spreads very quickly, so your first priority is to warn any people at risk. If in a building, activate the nearest fire alarm, call for emergency help, then leave the building. However, if doing this delays your escape, make the call when you are out of the building. As a first aider, try to keep everyone calm. Encourage and assist people to evacuate the area. When arriving at an incident involving fire, stop, observe, think: do not enter the area. A minor fire can escalate in minutes to a serious blaze. Call for emergency help and wait for it to arrive.

The Elements of Fire

A fire needs three components to start and maintain it: ignition (a spark or flame); a source of fuel (gasoline, wood, or fabric); and oxygen (air). Removing one of these elements can break this “triangle of fire.”

- Remove combustible materials, such as paper or cardboard, from the path of a fire, because they can fuel the flames.
- Cut off a fire’s oxygen supply by shutting a door on a fire or smothering the flames with a fire blanket. This will cause the fire to suffocate and go out.
- Turn off a car’s ignition, or switch off the gasoline supply.

Leaving the Building

If you see or suspect a fire in a building, activate the first fire alarm you see. Try to help people out of the building without putting yourself at risk. Close doors behind you to help prevent the fire from spreading. If you are in a public building, use the fire exits and look for assembly points outside. You should already know the evacuation procedure at your building. If, however, you are visiting other premises you are not familiar with, follow the signs for escape routes and obey any instructions given by the fire marshals.

Clothing on Fire

Always follow this procedure: Stop, Drop, and Roll.

- Stop the casualty from panicking, running around, or going outside; any movement or breeze will fan the flames.
- Drop the casualty to the ground. If possible, wrap him tightly in a fire blanket, or heavy fabric such as a coat, curtain, blanket (not a nylon blanket or an open weave type of any material—acrylic, wool, cotton, or other), or rug.
- Roll the casualty along the ground until the flames have been smothered. Treat any burns: help the casualty lie down with the burned side uppermost, and cool the burn by applying cool water or fanning the area gently.

Smoke and Fumes

Any fire in a confined space creates a highly dangerous atmosphere that is low in oxygen and may also be polluted by carbon monoxide and other toxic fumes. Never enter a smoke- or fume-filled building or open a door leading to a fire. Let the emergency services do this.

- If you are trapped in a burning building, if possible go into a room at the front of the building with a window and shut the door. Block gaps under the door by placing a rug or similar heavy fabric across the bottom of the door to minimize smoke. Open the window and shout for help.
- Stay low if you have to cross a smoke-filled room: air is clearest at floor level.
- If escaping through a high window, climb out backward feet first and lower yourself to the full length of your arms before dropping down.

Severe Burns and Scalds

Take great care when treating thermal burns that are deep or extensive. The longer the burning continues, the more severe the injury will be. If the casualty has been injured in a fire, assume that smoke or hot air has also affected his airway and/or lungs. Your priorities are to cool the burn (which stops the burning process and relieves the pain) and to monitor his breathing. A casualty with a severe burn or scald injury will almost certainly be suffering from shock because of the fluid loss and will need urgent hospital treatment. The possibility of no accidental injury must always be considered, no matter what the age of the casualty. Keep an accurate record of

what has happened and any treatment you have given. If you have to remove or cut away clothing, keep it in case of future investigation.

1. Start cooling the burn as soon as possible after the injury occurred. Flood the burn with plenty of cool tap water, but do not delay removal of the casualty to the hospital. Help the casualty sit or lie down. If possible, try to prevent the burned area from coming into contact with the ground.
2. Call for emergency help; if possible, get someone to do this while you cool the burn.
3. Continue cooling the affected area for at least ten minutes, or until the pain is relieved. Watch for signs of breathing difficulty. Do not overcool the casualty because you may lower the body temperature to a dangerous level. This is a particular hazard for babies and elderly people.
4. Do not touch or otherwise interfere with the burn. Gently remove rings, watches, belts, shoes, and burned or smoldering clothing before the tissues begin to swell. A helper can do this while you are cooling the burn. Do not remove clothing that is stuck to the burn.
5. When the burn is cooled, cover the injured area with plastic wrap to protect it from infection. Discard the first two turns from the roll and then wrap it lengthwise over the burn. A clean plastic bag can be used to cover a hand or foot; secure it with a bandage or adhesive tape applied over the plastic, not the damaged skin. If there is no plastic wrap available, use a sterile nonstick dressing, or improvise with gauze. Apply any dressing very loosely.
6. Reassure the casualty and treat him for shock if necessary. Record details of the casualty's injuries. Monitor and record his vital signs—level of response, breathing, and pulse while waiting for emergency help to arrive.

Minor Burns and Scalds.

Small, superficial burns and scalds are often due to domestic incidents, such as touching a hot iron or oven rack. Most minor burns can be treated successfully by first aid and will heal naturally. However, you should advise the casualty to seek medical advice if you are at all concerned about the severity of the injury. After a burn, blisters may form. These thin “bubbles” are caused by tissue fluid leaking into the burned area just beneath the skin's surface. You should never break a blister caused by a burn because you may introduce infection into the wound.

1. Flood the injured part with cold water for at least ten minutes or until the pain is relieved. If water is not available, any cold, harmless liquid, such as milk, can be used.
2. Gently remove any jewelry, watches, belts, or constricting clothing from the injured area before it begins to swell.
3. When the burn is cooled, cover it with plastic wrap or place a clean plastic bag over a foot or hand. Apply the wrap lengthwise over the burn, not around the limb because the tissues swell. If you do not have plastic wrap or a plastic bag, use a sterile dressing or a gauze pad, bandaged loosely in place.
4. Take or send the casualty to the hospital if the casualty is a child, or if you are in any doubt about the casualty's condition.

SPINAL INJURY

Injuries to the spine can involve one or more parts of the back and/or neck: the bones (vertebrae), the disks of tissue that separate the vertebrae, the surrounding muscles, and ligaments, or the spinal cord and the nerves that branch off from it.

The most serious risk associated with spinal injury is damage to the spinal cord. Such damage can cause loss of power and/or sensation below the injured area. The spinal cord or nerve roots can suffer temporary damage if they are pinched by displaced or dislocated disks, or by fragments of broken bone. If the cord is partly or completely severed, damage may be permanent.

Any of the following incidents should alert you to a possible spinal injury:

- Falling from a height, such as a ladder.
- Falling awkwardly, for instance, while doing gymnastics or bouncing on a trampoline.
- Diving into a shallow pool and hitting the bottom.
- Falling from a horse or motorcycle.
- Football tackle or misplaced hit.
- Sudden deceleration in a motor vehicle.
- A heavy object falling across the back.